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Title

PAYLOAD-AWARE TWO-TIER UAV ROUTE PLANNING FOR LAST-MILE DELIVERY

REVIEW REPORT – I

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Abstract

In UAV systems for medical and emergency last-mile delivery, conflicting objectives must be balanced: higher flight risk is carried by shorter routes, more battery is drained by low-risk corridors, and the fastest paths are rarely achieved by energy-efficient routing. No single fixed trade-off serves all payload types, yet the same objective weights are applied by existing planners to every mission regardless of the payload. In existing multi-objective planners, including entropy-based, diversity-driven, and crowding-distance Pareto selection methods, objective weights are encoded as a fixed scalar before path search begins. No influence is exerted by payload properties such as fragility class, payload weight class, and delivery urgency over which Pareto-optimal route is chosen. When payloads change, weights must be manually re-tuned and planners re-run by operators, which is unworkable under field conditions. The proposed work is a Payload-Aware Two-Tier UAV Route Planning system, wherein Pareto front construction is separated from route selection and an explicit algebraic bridge is added between Tier 1 routing objectives and Tier 2 payload metadata. In Tier 1, a multi-objective A*/Dijkstra graph-search is executed over an OpenStreetMap airspace graph, producing a Pareto-optimal route set across distance, energy consumption, and flight risk. In Tier 2, four operator-assigned payload tags are read and objective weights derived: w_{risk} is scaled by fragility class, w_{distance} by delivery urgency, and w_{energy} by payload weight class. Every Pareto point is scored by these weights via a utility function, and the argmax is selected as the mission route. Evaluation will be conducted on synthetic graphs (50, 200, 500 nodes) against single-objective A* and MOPSO baselines using hypervolume, spread, and Tier 2 accuracy. Hardware validation will use a Pixhawk quadcopter via ArduPilot SITL simulation and outdoor flights. Planned experiments will quantify Tier 1 gains over both baselines, evaluating whether non-extreme Pareto points are selected for mixed-priority payloads. The principal contribution is a formula-driven mapping from metadata to routing weights, linking mission context directly to Pareto selection without expert intervention. Route choice is set by payload properties rather than context-blind diversity criteria. The Pareto front is rebuilt periodically, serving all missions between updates.

1. Introduction

Unmanned aerial vehicles (UAVs) are a useful platform for last-mile delivery in scenarios where ground transport is impractical: medical supply chains to remote clinics, emergency drug drops during disaster response, and fragile diagnostic sample transport where road vibration risks contamination. In each scenario the UAV must fly a route from a dispatch hub to a delivery point across an airspace that contains obstacles, regulated no-fly zones, and variable wind corridors. A good route balances three conflicting objectives: total flight distance, battery energy use, and exposure to high-risk segments.

This three-way conflict is exactly what multi-objective UAV path planning is built to handle. Rather than collapsing distance, energy, and risk into one score before the search runs, a multi-objective planner keeps all three objectives separate through the search and returns a Pareto-optimal front: a set of routes where none can be improved on one objective without giving up ground on another [1]. The front tells the operator what trade-offs are available; it does not, by itself, say which route to fly.

One common way to sidestep the trade-off question is to form a weighted sum of the three objectives into a single scalar cost function and then run a conventional path planner such as A* on it. This scalar collapses the multi-objective problem into a single-objective one, and the operator adjusts the three weights to express the mission's priority before each flight. The approach works for a fixed mission type, but it fails immediately when the payload changes: emergency medicine requires a completely different weight profile from fragile biological samples or disaster-relief bulk supplies [2]. Manually re-tuning and re-running the planner between missions is operationally impractical in time-critical field conditions.

1.1 Background

Building the Pareto front, as introduced in Chapter 1, is well studied: the field has produced graph-search and population-based methods that generate well-spread, non-dominated route sets efficiently. The question that decides whether any such system is useful in practice is how the final route gets picked from the front. Chapter 1.4 reviews three directly relevant papers and finds the same answer in each: delivery context plays no role. None of them let what the UAV is carrying influence route choice, or even ask which route should be chosen for a given payload. That is the gap this project addresses.

Autonomous UAV systems for medical and emergency last-mile delivery work

under a constraint that separates them from general routing problems: the best route trade-off changes with the payload. A fragile diagnostic sample needs a low-vibration, low-turbulence corridor even at the cost of a much longer flight time. Emergency insulin needs the shortest path regardless of risk. Bulk disaster-relief supplies need the most energy-efficient route to maximise delivery range.

Payload-awareness itself is not absent from the UAV literature: variable payload mass has been used to adapt the flight-speed and energy-consumption *model itself* during trajectory planning, recomputing the underlying physical equations as mass changes across a multi-drop mission [3]. What is absent is any use of payload metadata to *select* a point on a pre-computed multi-objective Pareto front. Tier 2's payload weight tag is a discrete, ordinal class (light/medium/heavy) that is one of four metadata tags algebraically blended into a static selection weight once per mission; it never alters the per-edge energy model or triggers re-planning, unlike [3]'s continuous mass-driven physics adaptation. Tier 2 closes that gap by selecting among a fixed front from metadata, rather than adapting the model to continuous mass.

Real-world deployments make the operational cost of payload-blind routing concrete. In January and May 2019, Swiss Post–Matternet M2 drones carrying clinical laboratory specimens on fixed, payload-invariant routes crashed in Zürich; post-incident analysis confirmed that the aircraft flew identical corridors regardless of whether the cargo was a routine parcel or an irreplaceable blood specimen, with no routing adjustment for payload fragility [4]. Independent laboratory work explains why this matters: controlled studies on drone-transported medicines have shown that the vibration and turbulence encountered along UAV flight paths cause measurable payload degradation—insulin exhibits fibrillation risk, biologics undergo protein denaturation, and blood samples show elevated haemolysis indices when the route includes high-vibration segments [5, 6]. A comparative UAV blood-transport study confirmed that pre-analytic integrity varies significantly with flight-path characteristics, with routes of greater altitude variation and turbulence producing higher rates of sample rejection at the receiving laboratory [7]. In every case the root cause is the same: the route planner had no knowledge of what was being carried, so it could not prefer smoother, lower-risk corridors for fragile payloads or shortest paths for time-critical medicine. Operational deployments such as Zipline's blood-delivery network in Rwanda—which serves remote clinics up to 80 km away on a single battery charge—further illustrate the energy dimension of the gap: heavier blood-product payloads draw more hover and lift power and shorten effective range, yet the system plans the same fixed route regardless of payload weight, with no mechanism to select an energy-efficient detour for heavy cargo or a direct short route when an urgent, lighter payload makes that trade-off optimal [8].

1.2 Motivation

Three real operational scenarios motivate this work, and their routing requirements do not agree.

Fragile medical samples such as tissue biopsies and blood specimens are sensitive to vibration and to temperature. The routing priority is to minimise flight risk, taken here as time spent in turbulent corridors, sudden altitude changes, and wind-exposed segments. A route that adds 40% distance but halves the chance of vibration damage is the better one.

Emergency medicine such as insulin or antivenom has a strict time window. The priority is the shortest route, so transit time is lowest; flight risk and energy come second. For a 15-minute delivery window, the shortest path wins outright.

Disaster-relief bulk supplies such as water purification tablets and ration packs are heavier than medical payloads and are neither urgent nor fragile. The priority is the lowest battery drain, so a single charge covers as many deliveries as possible.

No existing UAV planner switches between these three routing strategies from a payload tag. The proposed two-tier system does this with no expert intervention and no replanning. In the UAV delivery literature, payload metadata (fragility class, payload weight class, delivery urgency) has never been the sole input to automatic Pareto point selection. That is the gap this project targets.

1.3 Objectives

1. To formulate multi-objective UAV route planning as a Pareto front precomputation task over a delivery airspace graph, and to distinguish the deterministic graph-search approach from stochastic population-based alternatives on theoretical grounds.

2. To design a payload metadata schema and a rule-based mapping that turns mission context (fragility class, delivery urgency, payload weight class) into Pareto point selection weights with no expert weight tuning.

3. To benchmark Tier 1 route quality and Pareto front coverage against single-objective A* and MOPSO baselines on identical delivery scenarios, using hypervolume, Pareto spread, and Inverted Generational Distance (IGD) as quality indicators.

4. To quantify Tier 2 selection accuracy by measuring how closely the selected route's characteristics match the payload tag requirements across mission types

(fragile medical sample, emergency medicine, disaster-relief supplies).

5. To establish the system's novelty: payload metadata is the sole input to Pareto point selection, in place of context-blind entropy and diversity criteria, through a deterministic algebraic mapping from payload description to route choice.

1.4 Scope

This project covers two phases. Phase I delivers a complete, validated software simulation system including the multi-objective A* graph-search engine (Tier 1), the payload-driven route selector (Tier 2), a MOPSO baseline, and a benchmarking suite evaluated on programmatically generated airspace graphs at three scales (50, 200, and 500 nodes). Phase II extends the validated software to physical hardware: a Pixhawk-equipped F450 quadcopter with a Raspberry Pi 5 companion computer, validated through ArduPilot SITL simulation and outdoor flight tests. The Tier 1 Pareto front is treated as valid for a bounded refresh window: it is rebuilt on a periodic cadence, or when environmental drift (e.g. a substantial wind-condition shift) exceeds a defined threshold. Missions that fall inside that window are served by Tier 2 against the current front, with no per-mission recompute. The project does not address multi-UAV coordination, dynamic obstacle avoidance during flight, or real-time path replanning.

2. Related Work

Multi-objective UAV path planning draws on both graph-search and evolutionary optimisation [1]. Within it, route selection from a computed Pareto front is usually guided by the crowding-distance rule introduced by NSGA-II [9], or by an entropy criterion that keeps the archive diverse while a manually weighted scalar sum makes the final pick; neither considers delivery context. Payload information, where it appears at all, is limited to continuous mass folded directly into the energy/speed model itself, not discrete metadata used to select among a pre-computed Pareto front. Three papers most directly relevant to the proposed work are reviewed below; Table 2.1 gives a structured comparison.

Table 2.1: Summary of related work

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
1	Chokkalingam & Ganesan (2026) [2]	Pareto-optimal particle swarm optimisation (POPSO) with dynamic parameter adaptation and Pareto–entropy archive selection to maintain front diversity during optimisation; the final single route is then read off via a weighted-sum scalarisation whose weights are set empirically per mission.	Entropy-guided archiving improves front diversity and adaptability to real-time environmental changes over static-weight baselines.	Final route choice still reduces to a manually re-tuned weighted sum per mission; payload metadata never incorporated, so fragile samples and emergency medicine receive the same hand-tuned selection procedure from the same front.

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Table 2.1 (continued)

S.No.	Author/Year	Proposed Method	Key Findings	Limitations
2	Tsoutsouras et al. (2020) [3]	Adapts flight speed and the energy-consumption model to the UAV's instantaneous payload mass during trajectory planning for multi-user delivery missions, updating the trajectory as mass changes between drop-offs.	Mass-aware trajectory adaptation reduces energy consumption relative to mass-agnostic planning across multi-stop delivery routes.	Uses continuous payload mass only as a physical parameter inside the energy/speed model itself; builds no multi-objective Pareto front and has no notion of discrete payload metadata (fragility class, payload-weight class, urgency) selecting among pre-computed routing trade-offs.
3	Ren et al. (2019) [10]	Crowding-distance-based NSGA-II (CDNSGA-II) for multi-objective UAV path planning, balancing distance and safety over an octree-represented urban environment.	CDNSGA-II finds well-spread, collision-free Pareto-optimal paths efficiently in octree-modelled urban airspace.	Crowding distance encodes only the geometric spread of the front; it cannot represent urgency, fragility, or payload weight, and mission context plays no role in selection.

2.1 Research Gap

Two of the three reviewed papers treat Pareto-front diversity as the central design concern, and only one of those two uses it to *select* a route: the crowding distance of [10] picks a single point directly from the front, while [2] uses entropy only to keep the

archive *diverse* during optimisation and falls back to a manually weighted scalar sum, re-tuned per mission by a human operator, to make the actual final pick. [3] does not build a Pareto front at all: it folds continuous payload mass into the energy model's equations of motion and adapts flight speed as mass changes across a multi-drop mission, but never selects among competing routing trade-offs by payload. Whether the mechanism is geometric diversity or continuous mass-driven physics adaptation, none of the three use payload weight, or any other payload property, as a discrete input to a Pareto-point *selection* mechanism; in [3], mass reshapes the cost model itself and never enters a selection formula. This is the gap the proposed system closes: Tier 2 treats payload weight as one of four ordinal metadata tags algebraically blended into a selection weight, and does not use it as a continuous physical parameter inside the cost model.

The gap is not an implementation detail; it is a design assumption shared by all three approaches. Entropy and crowding distance are both defined purely over the geometry of the Pareto front; mass-aware trajectory adaptation is defined purely over the physical energy model, with mass entering as a continuous parameter to that model, never as a selection criterion. None of these mechanisms has an interface through which discrete payload metadata, as opposed to continuous mass or point geometry, can enter a *selection* step. Adding payload information to them would mean replacing the selection criterion outright, which is what the proposed Tier 2 utility scoring does.

This work addresses three specific gaps.

Gap 1. Existing UAV multi-objective planners offer no payload-aware route selection: they apply a fixed geometric criterion independent of payload type, or no selection mechanism at all. Tier 2 maps four payload tags directly to Pareto point selection through a formula.

Gap 2. Existing planners do not re-use one front across mission types; they re-run the full multi-objective search whenever the mission context changes. The two-tier separation decouples front construction from mission arrivals: Tier 1 refreshes the front on a periodic cadence, independent of payload, and Tier 2 selects a route for any new payload tag against the current front with no re-run.

Gap 3. No algebraic bridge links a payload description to the routing trade-off it implies (e.g. fragility $\rightarrow$ minimise risk, urgency $\rightarrow$ minimise distance); the mapping has never been formalised. The weight-derivation formula supplies that bridge, and the mapping it defines is deterministic and reproducible.

3. Problem Statement

Existing UAV route planners either apply a fixed geometric selection criterion over the Pareto front, independent of payload type (Gap 1), or fold objectives into a single hand-tuned weighted-sum scalar before path search begins, which fixes the objective trade-off regardless of the payload. When the payload context changes, from fragile diagnostic samples that need vibration-minimal routing to time-critical emergency medicine that needs shortest-path routing, an expert must re-tune the planner weights by hand and re-run the search from scratch (Gap 2), because no algebraic mapping from payload description to routing trade-off exists to make that re-tuning unnecessary (Gap 3). In field conditions, that intervention is not feasible.

Formally, given: (i) an airspace graph $G = (V, E)$ with three-objective edge cost vectors $\mathbf{c}(e) = [d(e),\, \varepsilon(e),\, \rho(e)]$ representing geodesic distance, estimated energy consumption, and flight risk respectively; (ii) a Pareto-optimal front $\mathcal{P}$ of non-dominated routes pre-computed over G; and (iii) a payload metadata tuple $(F, W, U, P_{\text{lvl}})$ encoding fragility class $F \in \{1 \dots 5\}$, payload weight class $W \in \{1 \dots 3\}$, delivery urgency $U \in \{1 \dots 3\}$, and priority level $P_{\text{lvl}} \in \{1 \dots 5\}$, the problem is to select the route $p^* \in \mathcal{P}$ that best satisfies the routing requirements implied by the payload metadata, without expert re-tuning and without re-running the multi-objective search.

The inputs are measurable quantities the operator can assign from domain knowledge (e.g., "this vial is fragile: $F = 5$; delivery is non-urgent: $U = 1$"). The constraints are that selection must be deterministic, reproducible, and finish within one second of payload tag entry. The success criterion is that the selected route's characteristics (distance, energy, risk) match the payload's stated requirements, checked by a Tier 2 selection accuracy metric evaluated across multiple payload presets. Three conditions follow from the gaps. The selected route p^* must depend on $(F, W, U, P_{\text{lvl}})$, not on front geometry alone (Gap 1). The front $\mathcal{P}$ must be reused across missions between scheduled refreshes instead of rebuilt each time (Gap 2). The map from $(F, W, U, P_{\text{lvl}})$ to p^* must be an explicit algebraic formula, not an expert-set rule or a lookup table (Gap 3).

4. Justification for the Problem Formulation

4.1 Need for the Proposed Work

The gap established in Chapter 1.4 carries a direct operational consequence. Manually re-tuning three objective weights and re-running a multi-objective search between

every dispatch requires an expert on hand at the point of launch, which is unavailable in a time-critical emergency deployment. A system that requires this intervention is unusable in the field regardless of its routing quality on paper.

This is what motivates the split between Pareto front construction and payload-driven selection. The front is built offline for a delivery region and refreshed on a periodic cadence, or when environmental drift (e.g. a substantial wind-condition shift) exceeds a defined threshold, instead of once per mission. A mission that arrives inside that refresh window runs only Tier 2: it reads the payload tag and picks a route from the current front in well under a second. No expert, no per-mission replanning, no delay between tagging the payload and generating the mission file.

4.2 Feasibility

The work is feasible on technical grounds. The Tier 1 multi-objective A*/Dijkstra algorithm is a well-studied graph-search extension, and the Tier 2 utility scoring is a closed-form formula that runs in microseconds on any hardware. Both are implementable in Python with standard open-source libraries (NetworkX [11], NumPy). A Raspberry Pi 5 companion computer can host Tier 2 route selection, which runs end-to-end in under a second from payload tag input to mission file output.

Operationally, the system asks the operator for only four integer tags and needs no weight-tuning expertise. The ArduPilot SITL environment allows full end-to-end validation before any outdoor flight, so hardware risk stays low.

The cost is modest. All software uses open-source libraries and protocols (osmnx [12], NetworkX [11], NumPy, matplotlib, MAVLink [13]), and the hardware (F450 frame, Pixhawk flight controller, Raspberry Pi 5) is standard and commercially available. No proprietary licences are required.

The schedule fits the two semesters. Phase I (software simulation and benchmarking) fits the first semester up to Review I; Phase II (hardware integration and outdoor testing) runs through the second semester, with milestones set against the Review II and Final Review dates in Chapter 5.2.4.

4.3 Expected Contribution

Pareto-optimal graph-search is an established technique; the contribution is the bridge between Tier 1's routing objectives and Tier 2's payload metadata. That bridge is algebraic, not a lookup table or an expert-set rule. Because every weight depends on all three raw scores (Chapter 4.3), the utility function can select any point on the front,

not only its extremes, with no human deciding which extreme fits which payload. No prior UAV delivery system gives payload metadata this role; existing planners treat route selection as a front-diversity problem that ignores the payload. Once built, one front covers every mission profile in a delivery region until the next refresh, and route choice becomes a sub-second lookup from metadata.

5. Methodology

The system rests on one design decision: separate Pareto front construction (Tier 1) from route selection (Tier 2), for the reasons in Chapter 2.1. The front is built once per delivery region and refreshed periodically, not per mission. A mission that arrives between refreshes runs only Tier 2, which reads the payload tag and returns a route from the current front in a fraction of a second.

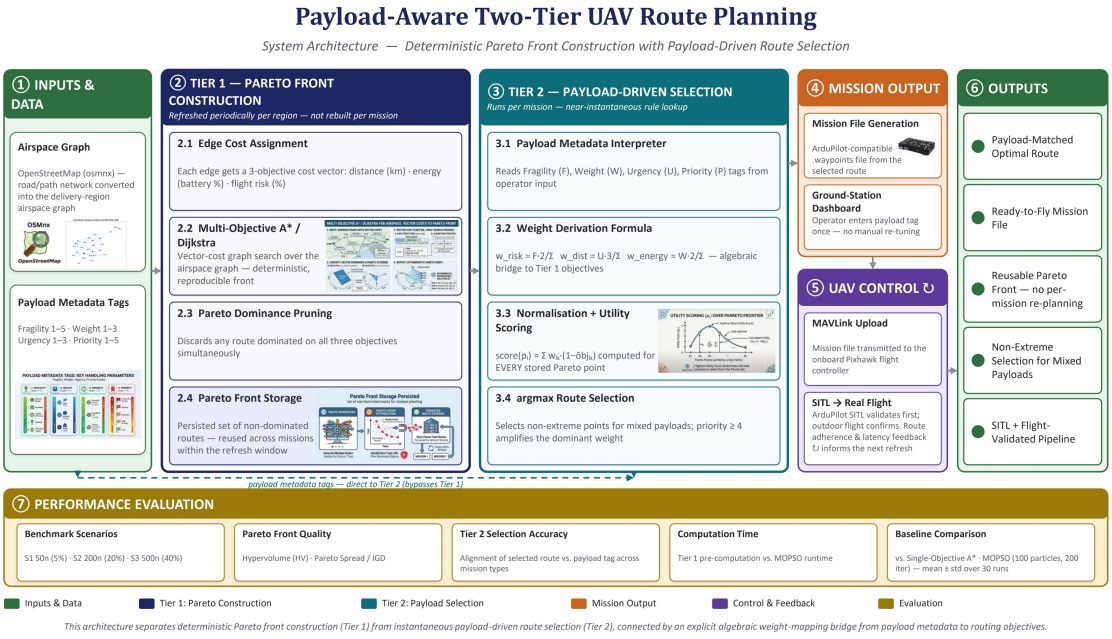


Figure 5.1: Methodology overview: Tier 1 Pareto front construction feeding Tier 2 payload-aware route selection.

5.1 System Architecture

Figure 5.2 shows the complete two-tier system architecture. Airspace graph data from OpenStreetMap enters Tier 1, which performs a multi-objective A*/Dijkstra search and stores the resulting Pareto-optimal route set. For each new mission, Tier 2 reads the four operator-assigned payload tags, derives objective weights algebraically, scores every stored Pareto point via a utility function, and outputs the arg max as the selected mission route along with an ArduPilot-compatible `.waypoints` file.

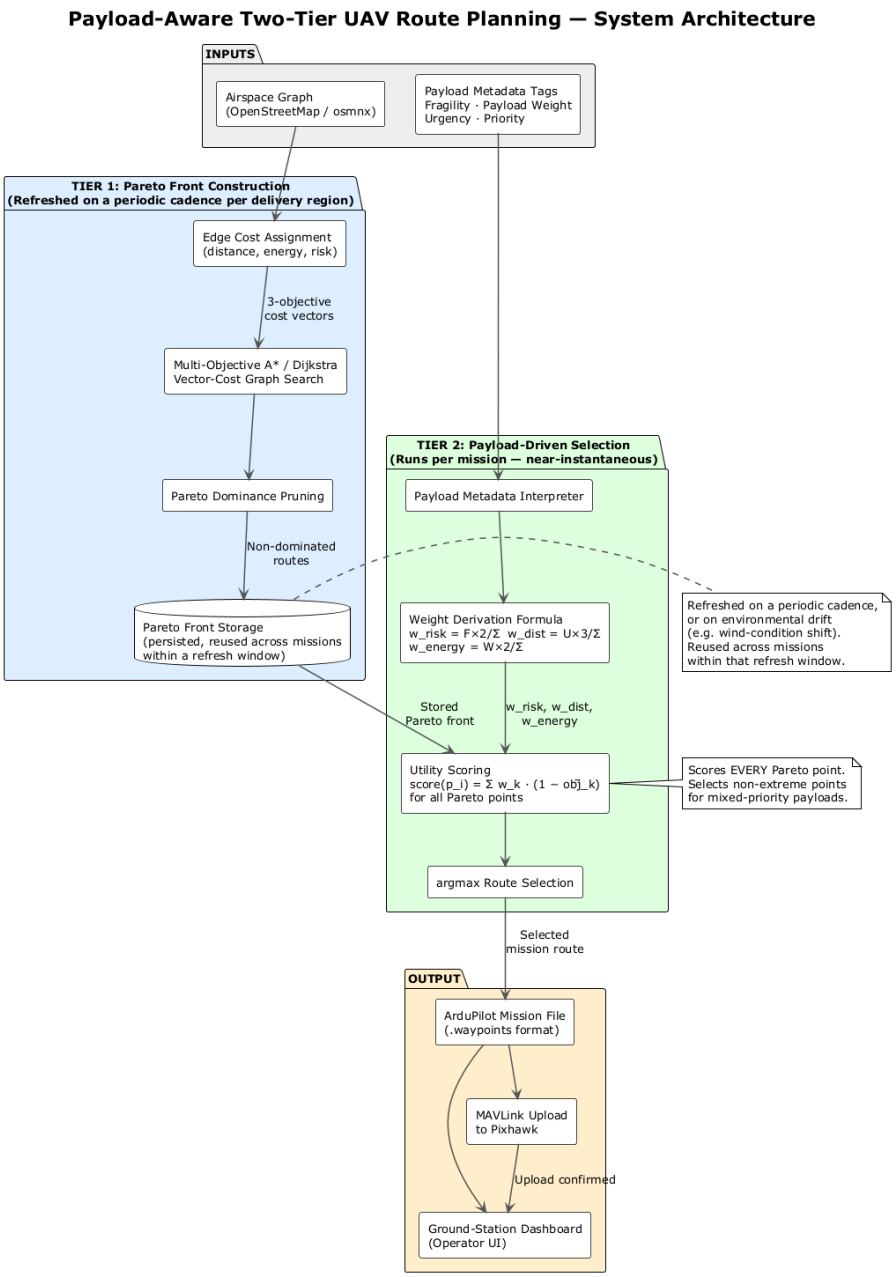


Figure 5.2: Overall system architecture: Two-Tier Payload-Aware UAV Route Planner. Tier 1 builds the Pareto front per delivery region on a periodic refresh cadence; Tier 2 selects the optimal route for each new payload tag using the current front.

The airspace graph $G = (V, E)$ is extracted from OpenStreetMap using the `osmnx` Python library [12]. Each edge $e \in E$ carries a cost vector:

$$\mathbf{c}(e) = \Big[d(e),\ \varepsilon(e),\ \rho(e)\Big]$$

where $d(e)$ is the geodesic distance of the flight segment (km), $\varepsilon(e)$ is the estimated energy consumption (proxy: $d(e)\times$ altitude-change factor), and $\rho(e) \in [0, 1]$ is the flight risk score derived from obstacle density, proximity to no-fly zones, and wind

exposure (simulated programmatically for reproducible benchmarking).

The three optimisation objectives are: (1) minimise route distance d; (2) minimise energy consumption ε; and (3) minimise flight risk ρ. A route p dominates route q if p is at least as good as q on all three objectives and strictly better on at least one. The Pareto front $\mathcal{P}$ is the set of all non-dominated routes.

The Tier 2 algebraic bridge maps each of the four payload metadata tags to exactly one Tier 1 routing objective, as shown in Table 5.1.

Table 5.1: Semantic mapping: payload metadata tags to Tier 1 routing objectives

Payload Tag	**Range**	**Maps to Tier 1 Objective**	**Reasoning**
Fragility class F	1–5	w_{risk} (flight risk weight)	Fragile items need low-vibration, low-risk corridors
Payload weight class W	1–3	w_{energy} (energy weight)	Heavier payload increases hover/lift power draw $\Rightarrow$ higher energy cost
Delivery urgency U	1–3	w_{dist} (distance weight)	Time-critical deliveries need the shortest path
Priority level P_{lvl}	1–5	Tie-breaker amplifier	Scales the dominant weight when scores are close

From the payload tags, three routing objective weights are derived algorithmically with no expert input:

$$r_{\text{raw}} = F \times 2, \quad d_{\text{raw}} = U \times 3, \quad e_{\text{raw}} = W \times 2 \tag{5.1}$$

$$w_{\text{risk}} = \frac{r_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{dist}} = \frac{d_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}}, \quad w_{\text{energy}} = \frac{e_{\text{raw}}}{r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}} \tag{5.2}$$

The multipliers (2 for fragility and payload weight, 3 for urgency) encode the assumption that urgency is more penalty-sensitive than fragility or payload weight in the UAV medical delivery context. Payload weight is tied to the energy objective because a heavier payload directly increases hover and lift power draw; no such direct physical link holds for risk or distance.

For each Pareto point p_i with objective values $(d_i, \varepsilon_i, \rho_i)$, min-max normalised

values $\tilde{d}_i$, $\tilde{\varepsilon}_i$, $\tilde{\rho}_i$ are computed over the full Pareto front $\mathcal{P}$. The utility score is:

$$\text{score}(p_i) = w_{\text{dist}} \cdot (1 - \tilde{d}_i) + w_{\text{energy}} \cdot (1 - \tilde{\varepsilon}_i) + w_{\text{risk}} \cdot (1 - \tilde{\rho}_i) \tag{5.3}$$

The selected mission route is $p^* = \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$. For high-priority missions ($P_{\text{lvl}} \geq 4$), the dominant weight is amplified by $0.1 \times w_{\max}$ to sharpen the selection.

The Tier 1 and Tier 2 activity flows are shown in Figures 5.3 and 5.4 respectively.

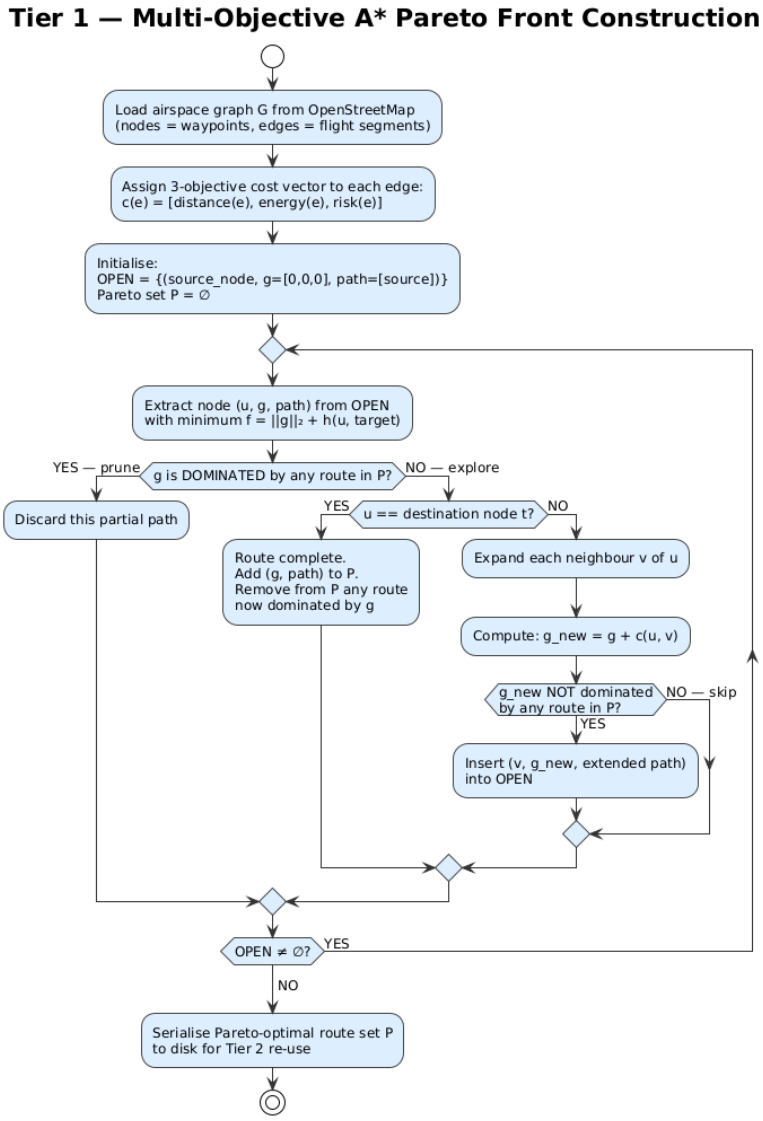


Figure 5.3: Tier 1 activity flow: Multi-Objective A* Pareto Front Construction. The open set is explored in order of estimated total cost until no further non-dominated routes exist.

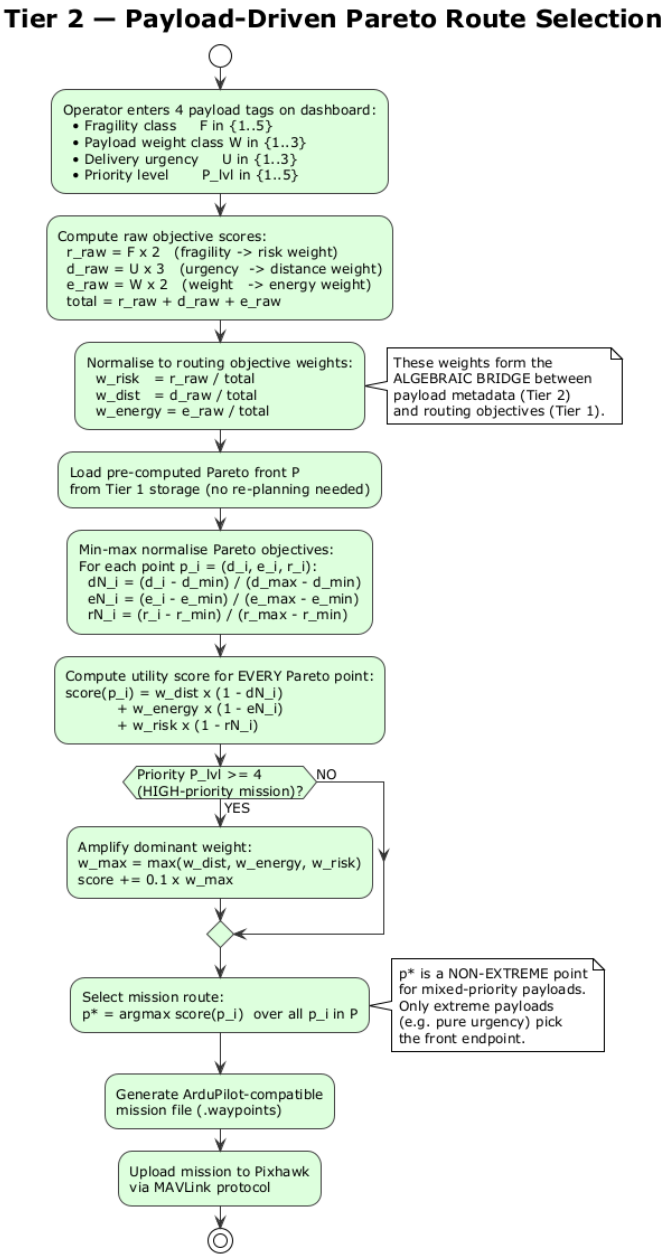


Figure 5.4: Tier 2 activity flow: Payload-Driven Pareto Selection. Payload tags are algebraically mapped to weights, which score every Pareto point via a utility function; the arg max is selected as the mission route.

5.2 Modules

5.2.1 Module 1: Airspace Graph Builder

Extracts the road/path network from OpenStreetMap for a specified delivery region using `osmnx`. Assigns three objective costs to each edge: geodesic distance, energy (distance × altitude-change factor), and flight risk (simulated threat zones and obstacle

density scored 0–1 using a configurable hazard map). Output: a directed weighted graph G stored as a serialised NetworkX object ready for Tier 1 search.

5.2.2 Module 2: Multi-Objective A*/Dijkstra Engine (Tier 1)

Implements the multi-objective A* algorithm maintaining a cost vector $\mathbf{g}$ (not a scalar) for each partial path. Paths dominated by already-discovered routes are pruned early. The `UpdateParetoFront` subroutine adds a new route and removes any previously stored routes it dominates. A node-level label-dominance check (NAMOA*) additionally discards any partial path whose cost vector is already dominated by a previously seen label at the same node, before it enters the open set, which avoids the exponential blow-up of naive MO-A* on dense graphs. Output: a deterministic, reproducible Pareto front $\mathcal{P}$ serialised to disk for Tier 2 re-use across all future missions in the same delivery region.

Algorithm 1 NAMOA* – Multi-Objective A*/Dijkstra Pareto Front Construction (Tier 1)

Require: Airspace graph $G = (V, E)$, source s, target t, edge cost $c(u, v) = [d, \varepsilon, \rho]$
Ensure: Pareto-optimal route set $\mathcal{P}$

$\mathcal{P} \leftarrow \emptyset$; $\quad labels[v] \leftarrow \emptyset$ for all $v \in V$
$g_0 \leftarrow [0, 0, 0]$; $\quad$ push $(f(g_0, s), g_0, s, \{s\})$ onto priority queue *Open*
$labels[s] \leftarrow \{g_0\}$
while *Open* $\neq \emptyset$ **do**
  $(f, g, u, visited) \leftarrow$ PopMin(*Open*)
  if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g$ **then** ▷ dominated by global front
    continue
  end if
  if $u = t$ **then**
    UpdateParetoFront($\mathcal{P}, g, \text{path}(u)$)
    continue
  end if
  for each successor v of u with $v \notin visited$ **do**
    $g_{\text{new}} \leftarrow g + c(u, v)$
    if $\exists\, \ell \in labels[v]$ such that $\ell \prec g_{\text{new}}$ **then**
      continue ▷ node-level label dominance – NAMOA* pruning
    end if
    $labels[v] \leftarrow \{\ell \in labels[v] : g_{\text{new}} \not\prec \ell\} \cup \{g_{\text{new}}\}$
    if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g_{\text{new}}$ **then**
      continue
    end if
    $f_{\text{new}} \leftarrow \|g_{\text{new}}\| + h(v, t)$
    push $(f_{\text{new}}, g_{\text{new}}, v, visited \cup \{v\})$ onto *Open*
  end for
end while
return $\mathcal{P}$

procedure UpdateParetoFront($\mathcal{P}, g_{\text{new}}, \text{path}_{\text{new}}$)
  if $\exists\, g' \in \mathcal{P}$ such that $g' \prec g_{\text{new}}$ **then return false**
  end if
  $\mathcal{P} \leftarrow \{(g, p) \in \mathcal{P} : g_{\text{new}} \not\prec g\}$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{(g_{\text{new}}, \text{path}_{\text{new}})\}$
  return true
end procedure

5.2.3 Module 3: Payload Metadata Interpreter (Tier 2)

Implements the payload-driven route selection algorithm: reads the four payload tags, derives objective weights via Equations (5.1)–(5.2), applies min-max normalisation to Pareto objectives, computes a utility score for every Pareto point via Equation (5.3), and returns the arg max route. Optionally amplifies the dominant weight for high-priority missions ($P_{\text{lvl}} \geq 4$). Generates an ArduPilot-compatible `.waypoints` mission file for upload.

Algorithm 2 Payload-Driven Pareto Route Selection (Tier 2)

Require: Pareto front $\mathcal{P} = \{(d_i, \varepsilon_i, \rho_i)\}$, payload tags F, W, U, P_{lvl}
Ensure: Selected mission route p^*

$r_{\text{raw}} \leftarrow F \times 2$; $\quad d_{\text{raw}} \leftarrow U \times 3$; $\quad e_{\text{raw}} \leftarrow W \times 2$
$total \leftarrow r_{\text{raw}} + d_{\text{raw}} + e_{\text{raw}}$
$w_{\text{risk}} \leftarrow r_{\text{raw}}/total$; $\quad w_{\text{dist}} \leftarrow d_{\text{raw}}/total$; $\quad w_{\text{energy}} \leftarrow e_{\text{raw}}/total$
if $P_{\text{lvl}} \geq 4$ **then** ▷ high-priority amplification
    $w_{\max} \leftarrow \max(w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}})$
    add $0.1 \times w_{\max}$ to whichever of $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ equals $w_{\max}$
    renormalise $w_{\text{risk}}, w_{\text{dist}}, w_{\text{energy}}$ so they sum to 1
end if
compute $d_{\min}, d_{\max}, \varepsilon_{\min}, \varepsilon_{\max}, \rho_{\min}, \rho_{\max}$ over $\mathcal{P}$
for each $p_i = (d_i, \varepsilon_i, \rho_i) \in \mathcal{P}$ **do**
    $\tilde{d}_i \leftarrow (d_i - d_{\min})/(d_{\max} - d_{\min})$ ▷ 0 if $d_{\max} = d_{\min}$
    $\tilde{\varepsilon}_i \leftarrow (\varepsilon_i - \varepsilon_{\min})/(\varepsilon_{\max} - \varepsilon_{\min})$
    $\tilde{\rho}_i \leftarrow (\rho_i - \rho_{\min})/(\rho_{\max} - \rho_{\min})$
    $\text{score}(p_i) \leftarrow w_{\text{dist}}(1 - \tilde{d}_i) + w_{\text{energy}}(1 - \tilde{\varepsilon}_i) + w_{\text{risk}}(1 - \tilde{\rho}_i)$
end for
$p^* \leftarrow \arg\max_{p_i \in \mathcal{P}} \text{score}(p_i)$
return p^*

5.2.4 Module 4: Testing and Evaluation

Evaluates all methods on three scenario scales: S1 (≈50 nodes, low obstacle density), S2 (≈200 nodes, medium density), and S3 (≈500 nodes, high density). The MOPSO baseline [14] runs with swarm size 100, 200 iterations, and a Pareto archive of 100 solutions, repeated 30 times per scenario to obtain mean ± standard deviation. Metrics include Hypervolume (HV), Pareto spread (the standard Δ metric, capturing extent and uniformity of the front), Inverted Generational Distance (IGD) computed against a reference front formed from the union of all non-dominated solutions found across

every method and repetition for each scenario, and Tier 2 selection accuracy across four payload presets. Phase II testing extends to ArduPilot SITL validation and outdoor flight tests on an F450 quadcopter with Pixhawk flight controller, measuring route adherence (GPS position error vs. planned route) and end-to-end latency.

6. Project Timeline Chart

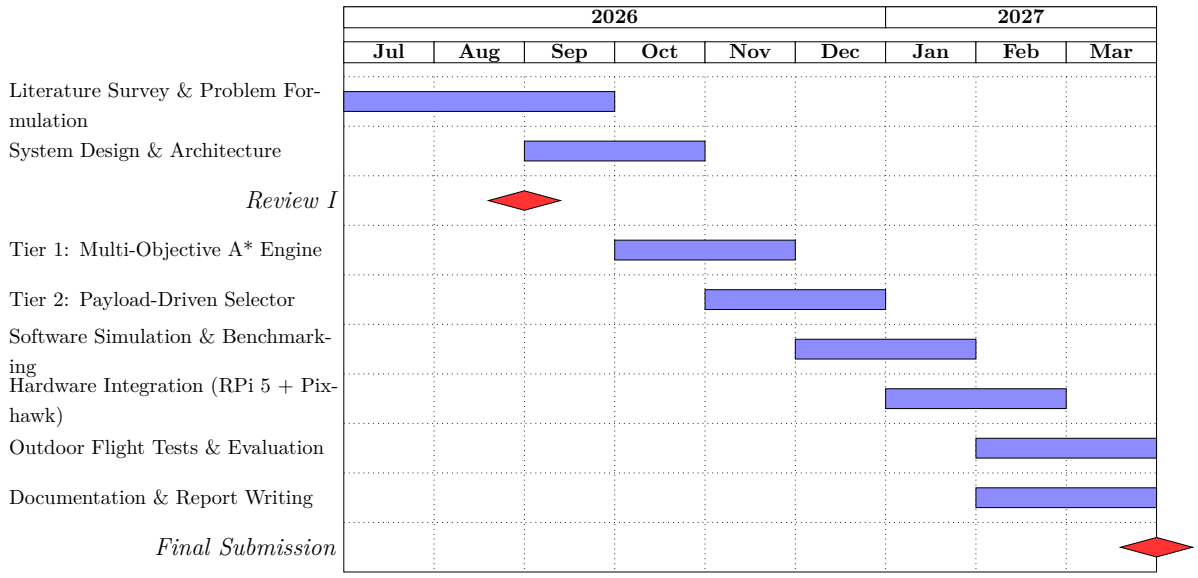


Figure 6.1: Project timeline (July 2026 – March 2027). Red diamonds mark review and submission milestones.

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